

ActInf Livestream #010.2: "A variational approach to scripts" (2020)

Livestream | 1:22:46

hello everyone welcome to the active inference live stream this is active inference live stream

10.2 it is December 15 2020 and we are really excited to be having this discussion with all of you

today welcome to the active inference lab everyone we are an experiment in online team communication learning and practice related to active inference you can find us at our website on Twitter at email our YouTube channel or our public qbase team and username this is a recorded and an archived live stream so please provide us with feedback so that we can improve on our work all backgrounds and perspectives are welcome here and as far as video etiquette for live streams remember to mute if there's noise in your background raise your hands so that we can hear from everyone on the stack and use respectful speech Behavior just as a quick little announcement SLO of process all the rest of the meetings for the rest of 2020 are at 7:30 to 9:00 a.m. PST last week we had 10.1 on the scripts paper and today we're having a follow-up discussion on the scripts paper

10.2 and then for the final two weeks of December we're going to be reading the paper sophisticated effective inference so check out the dates and um contact us if you want to participate in these final discussions or in 2021 or if you

have an idea for an event that we could do on the actm stream we have a lot to announce coming up soon and one good place to stay apprised of that would be on our website where you can find this call for collaboration link so if you go to active inference dorg and check out this aweber link you'll see some information about collaborating in 2021 on the live stream and on other really cool

projects great so today in ACM 10.2 we are going to start with introductions and warm-ups and then we're going to go through 10.2 we're going to be talking about the variational approach to scripts paper that we've been talking about for the last week as well and check out 10.0 if you want some context and check out 10.1 for last week's awesome discussion we'll go through the Asim claims abstract and road map today and then there's some new sets of questions about types of scripts and how do scripts look and other questions and then we'll um look through the figures which is something that we brought up last time and next week there's going to be a different paper so get in touch if you want to participate all right thanks for that now on to the intros and warm-ups so for the introductions and warm-ups let's introduce oursel by giving a short introduction or check-in and then passing it to somebody who hasn't spoken I'm Daniel I'm a postto in California and I will pass it to Blue I am blue Knight I am an

independent research consultant based
out of New Mexico and I will pass it to
RJ my name is RJ cord uh I'm on the east
coast of the United States and I'm
mostly doing research and organizational
Dynamics intelligence analysis and
narrative Warfare and I'll pass it to
Shannon if Shannon

here we'll come back to Shannon come
back uh Alex

then uh thanks I'm Alex I'm located in
Moscow Russia and I'm a researcher in
systems management school and I pass it
to

Sasha hi I'm Sasha I'm uh a graduate
student also in California and I study
neuroscience and

development and I will pass it to

Steven hello I'm Steven I'm based in
Toronto in Canada I do a lot of work
with participatory theater and S of
participatory development workshops and
I'm doing a practice-based PhD at the
moment and I'm gonna pass it to

Ivan hello my name is Ivan uh I'm based
in Moscow I'm interested in act in for
in in case of system thinking and I pass
it to Shannon

Shannon cool well I think Shannon will
come back online whenever she's ready so
for now we can just continue on to the
warm-up questions for these questions
feel free just to raise your hand we're
happy to hear from anyone so one
question since many of you were here
last week is what is something that you
thought about differently after our
conversation last week and so while

people are raising their hand um I just really like this way of thinking about scripts and genres for our encounters and at the life trajectory level and at the sort of sub scene level what genre are we drawing our scripts from RJ uh so after last week's conversation uh what I ended up really thinking about is happens when we lose our scripts so it it seems like scripts are are um a framework for representing like the like how how we how we move through these complex social interactions um so what happens when say like in the Spanish Inquisition where it was a process by which they were actively trying to remove scripts that people had um what happens in the absence of them uh when we tear them away so that that's that's I've been thinking about that a lot the last week interesting question blue I've been thinking really a lot about um like the transparency of active inference versus like other types of like deep learning and how they're kind of a blackbox and really what RJ uh had said last week about how active inference is more like a blanket of black boxes I thought that that was pretty cool and I've been like kind of you know twirling that round in my head for for the last week cool and it's almost like um if the sentence Oh go ahead Sasha um uh kind of the opposite of RJ for the last week I've been thinking uh

what's not a script um and can there be
an absence of a script because um we're
we're just as apt to follow another set
of directions as we are um the cultural
set of directions that we get for
interactions so um that's been kind of
uh vexing because it's it's similar to
the problem with um thinking about free
energy and active inference that once
you see it then you wonder like what's
not in that category and
um how do we kind of draw the correct
boundaries around uh the definitions of
these terms because yeah once you see
scripts you really see it around around
you in every situation yeah to your
point there Sasha it's like if a
firsttime ever social experience or a
very ritualized one if the scripts
concept can include both of those as
well as the internalist and externalist
concept it's kind of like how many
grains of sand in a heap like how
reproducible or how strong does the
pattern of a social encounter have to be
before we're going to deem it scripts of
a certain kind
blue so to kind of to piggyback on what
you and Sasha both saying like I've been
thinking about like is is maybe
everything is a script like is a chair a
script really like I mean a chair has
like a set role and like a a thing we do
with it or it can you know take on a new
like script so like what what about all
of these like objects in our everyday
Society really like once you once you
see a script like what is not a script

like I I I have a hard time like you
know unfolding that well it's like the
wood is chairing you know the chair is a
noun but it's a it's a process and it
was it built and it will eventually not
be a chair and so while it's chairing we
can draw a blanket you know put a seat
cover on it and use it in the functional
interface of being a chair and so
especially when things become um
sufficiently compartmentalized or
interface so that they have a functional
way of dealing with them instead of just
getting lost in the becoming and
Unbecoming of chairs we can start to
think all right well yes that is
happening but also there's this stable
pattern that I can sit on that's why
we're calling it a chair
and so it's like this emergence of
reproducible function and pattern out of
continual becoming and Unbecoming and so
that is to the point with a lot of
discussions about the marov blanket or
the frisen blanket whatever it ends up
being in these questions about
ergodicity as well like does everything
by virtue of having inputs and outputs
is everything going to be under this
umbrella or is this kind of umbrella
only going to be over things that have
um input output relationship ship of
some type over some spatial or temporal
scale

Stephen I think this also made me think
about where sense making comes in um and
how this approach to scripts it kind of
it helps extend the active inference

work with deontic cues you know they talk about the traffic lights and a lot of other things in the models and this starts to extend these sort of top down cues um into narratives and stuff like that but there's still the the more existential kind of participatory sense making and the kind of existential sense making processes which are maybe um they're still open to sort of somehow integrate with this more scripted kind of way of making sense of the world so I think that's that's something that I'm interested in in in where do they start to sort of mesh or override each other in different ways cool all right let's go to the next warm-up

question how could a different idea of scripts relate to how we regulate the behavior of ourselves and others so depending on how somebody thinks about scripts how might they be likely to treat their own behavior differently and how might they be likely or willing to engage in different kinds of regulations of other people's behavior so let's just say that there's some prop in a script so something like a uh artifact or an object so just like to RJ's example about wanting to uproot a script so then you could say well you can't use this symbol or you can't use this combination of words or we're not allowing anything from this genre of script and we're going to know it because it's going to have this style and um so depending on how we formalize these scripts maybe in

order to regulate other people's
deployment of scripts we might take very
specific um subcomponents like a certain
set of words or an artifact and then say
well this is or isn't ruled in

RJ uh yeah so so on on this um uh one
one thing uh going going back to the
last question just you kind of uh uh
stole the words out of my mouth what I
wanted to say which was is a a uh is a
chair a script for wood um but uh I I
also want to note like on this is I I I
got to have a conversation yesterday
with Eric Sapp from public democracy
and a lot of the stuff they've been
doing uh as I mentioned to you before
Dan is like um using like traditionally
what what would be used for for
advertising uh for for for like public
good so uh what they've done in one case
was was be able to uh use the same tools
that they would use to Target
advertising to um uh Target recent
people who have very recently picked up
a heroin addiction um looking at their
their behaviors online be able to figure
out that that they're that they've
recently picked up this addiction or or
that they very likely had and then give
them on-ramps toward treatment at the
time when they are most likely um uh
able to to move toward accepting those
types of treatments they're they're not
they're not very deep into the addiction
yet so um I I brought up scripts to him
uh and we we talked a little bit about
it and it seems like their look at it is
like yeah it's it's almost like one if

if you can give people the idea um that
um they're using these scripts and that
they can change them and that they're
creating them in real time uh or or
adapting to them in real time then um
you kind of have the self-awareness that
can kind of help you um moving down like
to we we talked about radicalization
last week like if you're aware of of of
the scripts that are in the space may
maybe you can you can be self-aware when
you're using them uh so I think that's
one thing and then um in addition like
recognizing when you're about to hit a
threshold where you're about to like
start using a script rather than just
some Behavior which kind of ties to what
Sasha was saying is like at what point
do we call something a script so I'm not
sure but but anyway yep um it it reminds
me of the Gateway idea like gateway drug
or Gateway website or Gateway meme
whatever it is it's like using that
metaphor it's like whoa if you go
through the Gateway if you pass through
you're not coming back it's all downhill
from there well that's one script it's a
narrative for how you could proceed
through different types of experiences
but could there be a sort of like
visitation metaphor it's like hey you're
going to visit this landia and then your
family is still going to be waiting for
you and they all want you to become uh
safe and come home for example and then
that could be a attractor that takes
some of the same setup it's a similar
genre at least in the beginning of the

movie but then it goes a different way
and so that definitely is a nice
point there okay and then our last uh
warm-up question with something that
you're still wondering about or
something that you'd like to have
resolved by the end of today's
discussion I really like this question
also about the what's not a script or
defining by defining what isn't another
thing that wrote I wrote down to ask
about is the transparency of the
concept is something that we'll probably
return to many times and is it really
more transparent to subc
compartmentalize it into a lot of
smaller black boxes or are we doing
something quite distinct from that what
else does anyone think like what could
be a fun area to take this script's
concept in or what could be somewhere we
could go in the
discussion but if no thoughts here then
no worries because there's a bunch of
questions that will be asked in the
later stages so good okay Shannon I hope
that you can hear us or rejoin when you
can but let's get into it and then
anyone just raise your hand when things
come to you and we'll go from there cool
so the paper that we are discussing this
week as with last week is a variational
approach to Scripts and we had Mao last
week on the panel which was great the
aims and claims of the paper we went
through those in 10.0 and heard it from
the authors themselves last week but the
big question here is about unifying the

study of scripts using the active inference framework in all of these different dimensions the strong and the weak and the internal and the external and the social and the neurobehavioral all these different aspects of scripts how are we going to reframe it within the active inference framework and then what would that do what would that give us in terms of understanding or unique predictions or explanations in the abstract they also run through this notion that they're going to be leveraging the resources of active inference to offer up a formal model of scripts that's not just a flowchart or a mindmap or Pros but is actually a relationship amongst variables that specifies how some of these script uh variations like weak versus strong internal versus external can be considered so we'll return to that in the figures especially and then in the road map we saw that classic formula with an introduction and then Theory X in this case script Theory and then active inference in section three and then in the following section A summation of theory x with active inference so that's the active inference account of scripts and this was an introductory work so there's probably many ways to follow down the rabbit hole for a specific subtype of script or other dimensions but um the main dimensions of variability in the scripts concept that were highlighted by the authors were

this strong versus weak and internalist versus externalist so the strong scripts concept reflects event sequences that are extremely stereotyped or extremely reproducible or um kind of like that Grand Canyon in the landscape it's like you're being channeled along very strongly by a specific uh script whereas on the weak side we see things that are more probabilistic or potentially a preference or bias or more likely to go one way than another but it isn't exactly pre-specified and then on the y-axis we have this internalist versus externalist axis so the internalist perspective is going to look towards smaller things things that are inside of the person like the brain or the cells or computational things the cells are doing or something that the neurons are doing in terms of secretion or firing or electromagnetic rhythms whereas the externalist perspective as it just would sound like is thinking about things that are external to the focal agent so that would be the relationships in The Social Network as well as quote society and even broader things like the ecosystem cool so those were all things that we talked about in the previous weeks and now I want to raise some new questions about scripts relationships and active inference so the first question is how might scripts of different kinds whether the strong weak internal external or some other feature that people are

wondering about how might they
conceptualize of social relationships
and in this case just for reference I'm
thinking of this social network on the
right there's two blue nodes let's just
think of them as our Focus relationship
so now we're moving beyond the
individual as the focal system let's
think about the edge and the
relationship as the focal system and
then seeing that edge in context of a
broader Network so what might anybody
think about um how would scripts
conceptualize of social relationships
and then I'll just throw up the second
question too which is how could active
inference add depths and tools or
context to the social scripts concept
any thoughts from any domain are good
Stephen I might be curious like to hear
about from a practice side how do how do
we bring these ideas and then make them
like tangible and
experienced yeah I mean I I think this
is this is actually where this scale
question becomes kind of kind of it
becomes a challenge right because as we
try to M map things out um with social
network so there's a lot of social
network or participatory inference map
is one process which is quite useful for
looking at and trying to understand how
we um relate to each other's stories and
experiences in our network but the
question becomes is um how do we relate
this um this kind of perspective we try
to take on stuff with the scripts we use
and then come back to well how much of

those scripts
are just
manifestations of embodied inactive
ecological behavior and we've kind of
rified them so um I suppose um it it
becomes this we we we it's very much I
think there's a kind of a um a
situated um way that we we we kind of
reenact or recapitulate this whole thing
depending on the discussion so you know
like if you're all sitting around a
table like in that board meeting where
people are putting together pictures of
a puzzle
they might be working with the scripts
of how they should um assemble their
perspectives and then you've got someone
in a who's performing in a ritual and it
might be much more about how they need
to hold themselves as a person so I
think there's something quite
interesting that this whole scripts
thing raises in terms of of the scale
question and I suppose that's partly why
you know Maxwell sort of ended up
working a lot in here as well because I
think it ties back to the question of
what scale is the script happening at
and that has a big impact when you try
to then talk to people in a practice
context because if you're working with a
community or an organization or
strategic strategy levels doing
strategic foresight they they may be
having their own kind of um scripts that
shape their social practices in that
kind of domain so I think this is this
is kind of

interesting what you just said there reminds me of uh two people like looking at a river and they're seeing different things they're seeing different scripts for The River there's no disagreement about what the river is doing but one person looks at it and says yep at the fork it goes left and the other person says it goes with the flow and today it's going left and so in that sense um they're seeing different scripts but they're not disagreeing about what actually is happening and so it's like there's multiple Concepts that are being projected on to the observables and then there's the story about those two people and how they reconcile their perspectives by the riverbank and so yes the scripts concept does seem to apply everywhere um but maybe to uh begin to Chisel away at what isn't a script I would say a chair could be a prop but I don't think a chair has a model where it is actually in agent doesn't have agency it's not an active entity in the script though it can be mobile ized by actors in a way that a prop can be can become enlivened but it isn't with a generative model of that situation

RJ yeah I was going to say yeah the chair the chair is an object that can be a part of a script but but it I don't think it can it's not a script itself and it can't act out any scripts right because it's not it's not an actor um uh uh unless maybe you know going out there it's like maybe if it were a mechanical chair of some kind like a

message chair then it has been script
that has been encoded by by some other
actor um but yeah don't Sasha and then
Stephen okay um that was a plot twist
with the message chair uh but perhaps uh
the ability to um participate in a
script does that does that imply that um
you're able to respond differently in
different situations because uh a chair
will always be chairing
um but an individual or a participate
yeah participator in a script um they
can

change uh based on the context and the
kind of script that they're
participating in so maybe that's a key
aspect of uh being a participator in a
script

Stephen yeah I think this chair is quite
an interesting example and iose like in
theater they often work with mask
and The Mask I think they might be sort
of bringing this back to the bodies like
there's there's something about the mask
and identity and taking off a mask and I
haven't really thought about this before
but I don't know say a throne compared
to another chair could the throne almost
be coated with a mask you know I mean it
starts to have an identity and it's like
you know so I mean a red nose is
basically the smallest mask you can wear
is basically become a clown but there
all this work and we don't tend to do
much with masks but if you go back to
sort of indigenous cultures everyone was
working with masks you know um so
something interesting in is there

something about the script which is
about how we broadcast and receive
through this part of the
body thanks

RJ uh yeah no no no so so

um bouncing off what Stephen's saying um
so one one of the things when I first
read the paper I talked to you about
this like immediately after reading the
paper paper I think was um that uh if
you look at uh there there's a common
rituals between North American Indians
Nordic cultures African like all over
the place where where um what happens is
there's there's it's either dances
something along those lines where you
use animal masks and act out the scripts
that those animals use it's really
important especially for you know like
uh so I I brought up to you Dan at one
point the bear uh in particular because
bear is an animal that has a set of of
of scripts that you can identify that
it's using um and and if you don't have
experience um you know knowing like
reading their behavior then you're more
than likely going to get hurt or put
yourself in a really dangerous situation
so like um park services in North
America routinely make the point that
almost all negative bear encounters
happen because people don't know how to
interact with the bear they run um
they're they're not doing the right
Behavior so even with like a like a
black bear they play dead it's like
don't play don't play dead with a black
bear because they're going to

investigate they're want to figure out
like okay easy you know easy meal
they're they're they're they're they're
curious and they're going to go mess
with you um so what what these rituals
did was they gave young people who
hadn't had the experience yet an
opportunity to see those scripts in
action then before they may have ever
interacted with the animal they may be
asked to play out that script take the
place of the bear wear the mask and and
it gives you so acting out the script of
the other person gives you a better
sense of or or in this case the animal
gives you a better sense of what it's
like to be that animal so um in that
case like acting out those scripts that
other people may be using or you know
other other act other actors would be
using um you know actually is an
advantage in the environment um on you
brought up the throne though and it's
like yeah it's it's a um well I I like I
like the presentation of it as a mask
for a chair uh but uh it's it's really
important in denoting the person who
sits in this chair should be acting a
certain way right and and when they
don't it's a joke right so I don't know
I was just thinking about like like a
few things there there there's a it's
almost like there's a um a real
advantage to understanding and then
acting out the scripts of
others nice R.J the hero with a thousand
chaires as it were um I also didn't
bring up wolves and I wasn't praised for

that so exactly oh wait
blue so just kind of thinking about like
the chair itself is actually like
supposed to act a certain way like
you're supposed to do certain things
with the chair right and and I mean I
I'm always telling my kids like that's
for sitting you've got the wrong end of
you on the chair like so people get
creative right like with the different
roles that a chair can play it can be
like a step stool or it can be you know
some piece of gymnastic equipment and
it's just it's just interesting to think
about changing the script even for a
chair let alone the person that's
sitting in a chair
it's it it's like the setup for an
improv scene and yeah the different
kinds of chairs are going to lend
themselves to different scenes maybe
with a throne maybe they could enact a
classical monarchy script or maybe it's
a Monty Python it's a satire or with
certain kinds of chairs or in certain
context you can enact a shivalry script
and you can pull the chair out for your
colleague so there's all kinds of ways
that um different props facilitate or
enable different script to play out and
you can always do a mime like you can
sort of act out a uh digging or you can
act out playing baseball but you do need
the props our extended cognition to
actually do some of these scripts that
involve our extended Niche
Sasha um I just wanted to touch on
something Stephen said uh that um you

said today we don't do much with masks
um um I I guess I would disagree because
it seems like uh a lot of us are doing a
lot with masks and um just yeah kind of
obvious to think that um that's a prop
that is going to change our behavior in
ways we might not even recognize and um
yeah just important to think about
um how this uh new prop is changing
human interaction and scripts and um
behavior in ways that we don't want or
really expect uh a mask to
change

blue so Sasha didn't come right out and
say it but I'm gonna like maybe point it
at what you're saying Maybe not maybe I
took it out of context but um I I think
that we do do a lot with masks like we
are right now using the biggest mask
ever like people are totally different
in like their online Persona like some
people will hide behind a computer
screen and act in ways they would never
ever behave in real life right so they
have this like mask up they can create
this new persona and and go forward in
the world with

that yep very true and with new props
there's new scripts there's the one
person's wearing a mask but the other
one isn't script there's the is that
person wearing an n95 script there's all
these things that wouldn't have even
made sense in 2019 so that's the sense
making and the cultural sense making and
the tension and the friction when
there's differences on opinion or
differences in behavior and it's about

um not just estimates of state but actually uncertainty and estimates about other people's uncertainty that's this whole like recursive and embedded element of active inference and it's about navigating the uncertainty of a situation not just maximizing the state itself and that's something we'll return to for sure

Stephen yeah I think this is interesting when you start to put the mask so this's I suppose I was talking about you know playing with a mask physically within this proximity of your face but yeah there's there's this kind of mask out there but it in some ways it's become abstracted and um like it's it's almost like like you can you put it on and put it off it's it's it's sort of almost part of the ecosystem that we've kind of created which is maybe a little bit unhealthy because we don't um there's actually one really interesting activity I did once there was a chat from Rwanda who does healing work and he would get you to put a mask on of your face and read a poem that was really personal and then you had to lower the mask but it was like you were lowering the whole of your sort of masking identity and it left you feeling this incredible rawness which is really interesting so I suppose um yeah there is this tendency now that and I think even maybe more since the 1980s if I look back at videos of people in the 80s on TV and that they're often a lot more nerv in front of the camera now

people are used to putting up a some
sort of smiley face type of thing and
presenting I think there's um there
there's something quite interesting in
that that might relate to this paper
yeah I've been thinking about that too
like with the selfie new affordance of
being able to at an arms length look
back not just with a disposable camera
and snap the pick and have the
perspectival imagination of being at
your arms end but actually seeing it in
real real time and then video chatting
or live streaming where you can see your
own face during an interaction so it's
like you're seeing the mask in real time
and then that's sort of like um I think
it's a Campbell quote or could be
someone else but like uh if you think
that you're rejecting the mask that
Society gives you you're just wearing
the mask of rejection everybody can see
that it's not exactly the quote but
basically it's like you can't not play
the mask game there has to be an
epithelium there has to be a facade
there has to be an interface and so
we're talking about designing that
interface because somebody who goes
around saying No this is me this is me I
have no mask this is just what I am it's
like right but there still is an
external layer that's being presented
right and if someone says no I'm I'm All
me there's no external level it's like
wo big of true of course but then it
it's not a helpful Viewpoint for the
world because we do need to interface we

need to send emotions reliably and receive emotions reliably and so um lot there let's continue with some of these fun questions so the third question is um how do the network relationships the red nodes influence potentially indirectly the focal edges in the network blue and then the fourth question would be like little bit more of a wild card what could be a fun or interesting social experiment related to active inference whether it's something that we do like a little experiment or just daily routine tweak or whether this is something that could be carried out with a larger scale laboratory online or in a social setting um so either of these questions people could raise their hand on I would uh start with that third one and uh highlight that that edge it's a relation but it's like an atom in the world of relational thinking the edge is like the atom and so now we're in this network of edges which is what makes it kind of a tapestry and a social Fabric and so if we want to understand the relationship between person A and B then we're always invoking all the other edges that stabilize them whether it's the who brings the food to these different people or how do they get their internet because they're not going to be able to connect if they're not going to get their internet and what are the Norms that they're getting the other 23 hours of the day and that's what shapes and in some ways can even Define the focal relationship is again the

other 95% of the person's life or day
that they're not in that
relationship um during
RJ all right so um uh just turning hand
off there we go uh so on the last
question I think I have um something
I've been thinking about that may relate
to the prior One on network
relationships um so uh uh my dog
Achilles uh he usually wakes me up in
the morning uh for for like for last
year or so uh we we go on a walk at at
like 5:00 a.m. um and he is usually
raring ready to go on that walk so I
never really had to worry about an alarm
clock he always you know he would always
P me awake um and if I like there's no
snooze button on a dog it is you know he
he's there until I wake up um but
recently after Thanksgiving um so I have
Achilles and my parents have his brother
Socrates and Socrates has been staying
with us so now Socrates plays with him
during the day wears him the hell out
and uh ever since Thanksgiving um he
doesn't wake up in the morning anymore
he doesn't want to wake up he wants to
sleep in so he had a script that was
present in the morning that he now no
longer has and this has led to me
oversleeping almost every day because
I'm not I don't have any script for for
interacting with an alarm clock in the
morning anymore uh so uh what I think
that uh you know going right back to the
beginning of like what is not a script
and and where's the transparency on like
on on being able to look at look at what

these things actually look like I think
um our wakeup uh scripts is are are
probably some of the most transparent
because they're the most consistent um
and they're the most um uh you know kind
of there there's not a lot of uh outside
interaction with those scripts it's it's
it's uh very much like at the moment
when you're waking up becoming conscious
in the world so there's not a lot of
Prior of them maybe your dreams Etc um
so I think it would be kind of a fun
experiment to to have people try to
identify what it is their morning script
actually is um identify where their
first Network relationship is in that
script right because maybe it's I know
some people who wake up and immediately
go on their phone um uh or you know like
so so then um uh try to see if they
can't pick out one part of that script
that's not good for them that they don't
like that's that's not healthy um uh and
then like you know bringing back to my
question which I wanted to resolve last
last week was how do you identify the
threshold events where you're able to
change scripts I think it would be a fun
experiment to try to figure out like
like what are the best methods for
changing your morning scripts which are
definitely some of the most difficult to
change good questions RJ thank you that
um reminds me of somebody being asked
what is your morning routine or what's
your bedtime routine and I'm just
projecting but a common response could
be I don't have a morning routine it's

like well surely you do have a probabilistic distribution of places you wake up and how often you brush your teeth even if it's 0% but there's some distribution of routines there that's the script it could be a weak script it could be a strong script but some people just brush off the question no no I don't have a mask I don't have a routine it's like a way of not engaging with that but once you recognize that it's about the interface and it can be a probabilistic interface just to say a morning routine it doesn't need to be regimented it doesn't need to be externalist it it doesn't need to be internalist we're moving beyond internalism and externalism as well as Beyond um um restrictive strong or loose weak we're taking all of those possibilities and now we're just thinking yes so what is your morning script set or maybe what what genre are your mornings um is it an emergency movie is it a comedy and then also this question about the dependencies and to take it back to this social network and the scripts it's like we often don't know all the dependencies until we do a perturbation so like RJ you didn't know about your lack of an alarm clock you know stimuli or script until there was another script that changed in a non-human animal and then that had a cascading failure in this network of dependent scripts so it revealed new relationships and dependencies because of a manipulation and so that's very

related to active inference on a few levels just to provide one is we're always interested not just in coming to Clarity on world States but actually coming to Clarity on which interventions or which experiments or perturbations would maximally inform us while also remaining functional about those dependencies so then you could be um you even when the dog was still waking up early you could have been experimenting with earlier or later wakeups and maybe by uh bumping things by half an hour this way or another you would have discovered oh wait without the dog I don't have a way to wake up or if the dog is tired then there's this other relationship and nobody would connect the dog being tired to waking up early for most people especially if they don't have a dog but in this case it could have been explored or actively inferred with um rightful experimentation

Sasha um yeah that's really kind of funny about the morning routine because um yeah we we of course all have one but maybe it's uh not not very glamorous or uh effective and it just reminds me of um these sort of uh glamorized morning routines of celebrities where like they spend um you know all this time putting stuff on their face essentially and uh it kind of signals like I have the kind of morning routine that I'm proud to tell other people about or should be a you know a sensational YouTube video that other people Envy um and so kind of changing yeah what a morning routine is

or should be and then also a part of that whole like um kind of uh thinking about your morning routine is that it's very solitary when in fact of course we wake up for other people or other creatures and that um we kind of get sucked into someone else's script even before we know that we're part of it um like people who wake up to take care of their kids like maybe they don't want to be waking up but you know um and then how to use active inference in a way that um helps people modify their um morning script

um in a way that I I guess it just kind of reminds me of like biohacking and and um being very like uh focused on every metric that you have so do you have to have all those metrics in order then to change your morning script or like to optimize that or is there a way where active inference can help us get around that um kind of uh self-measurement Obsession that biohacking can fear off into um when it can be more focused on just

uh yeah the the performance and the human script that we're participating in good question there Sasha and it reminds me of just the imperatives in active inference being to reduce uncertainty and to design optimal experimentation and so instead of um in this reward framework thinking we're going to have the most rewarding morning thing it's going to set you up for the most productive day or it's going to feel the best it's all about optimization most

most most and here we're actually saying
yeah maybe we'll get there that'd be
great but just where we are how could we
reduce our uncertainty like maybe every
morning you wake up and you're not sure
whether you're going to take mechanism A
of commute or B so how could we start by
reducing uncertainty and decreasing
anxiety about the coming day increasing
Clarity and then asking not what is the
optimal state but what's the optimal
experiment and policy that's within our
scope of affordances not like waking up
and instantly going to the strange
attractor of thinking that something
could be a different way um we can focus
on our policies and then this Beyond
internalism and externalism it's just so
critical because it's uh sort of the
false dialectic of like worrying about
oneself only and then disregarding
systemic issues versus getting extremely
obsessed with systemic issues and then
losing the sight over the affordances
that we do have and in the middle um is
the compromised language but we're going
to not just be in the middle we're going
to be all over the map and we're going
to be able to dive into the internalist
thinking when we need it we're going to
be diving into the externalist thinking
when we need it but we're always going
to be able to return to the fullness of
the niche and what's possible and
getting other people's perspectives on
the issue so that's I think a a a market
difference in active inference um let's
uh go on to the next slide

so this one's just kind of a little fun
single question I was just wondering
what does a script look like and so the
movie script this is the Lethal Weapon 4
autographed Edition script um one of a
kind uh that's a movie script so the
words are syntactically encoded and
maybe the semantics are suggested or it
gives stage directions like a dramatic
script and then in the middle is this
like peer pressure script like what's
the script that keeps people driving on
the right side of the road in America
and on the left side of the road in the
UK that is something that's enforced by
Semitic cues like Signs and um
counterfactuals like if I were to drive
on the other side of the road I would be
in the uh felony script or something
like that but how does pure pressure
play out and how do we visualize it and
then just a big unknown is what do these
active inference scripts look like like
how would we communicate the scripts and
so instead of um using a movie dialogue
to convey some of these probabilistic
things or the peer pressure um narrative
to talk about some of these things is
there another visualization or
representation just I'm not sure if
anyone has thoughts yeah RJ or and then
anyone else go
ahead uh well I think um uh I I don't
know if um I'm pretty sure I'm pretty
sure the graphic was was in was in the
um was in the slides uh last time was
like uh kind of you know just just a you
know 2D representation of the axes of

like strong script to weak script
internalist versus externalist and um
you know the the paper itself like like
tried to show that it's like look these
these are just um you know these don't
have to be Divergent views they're just
different perspective yeah that's it yep um
and I feel like uh you know the the
actor script is just a a it's a
Continuum it's very strong externalist
script you're either performing The
Script exactly or it's not it's not it's
not being performed it's not happening
um then you have like yeah I brought up
Always Sunny last time it's like it's a
show that allows for a lot more
improvisation so the scripts they use
while strong and externalist aren't
as strong as say Lethal Weapon right um
but then peer pressure is something that
that's that's far more internalist your
your interpretation of what you
think the script is and maybe talking
with people about what you think the
script is and trying to work that out
and it's a weak script where it's like
you can even see there that like in in
the picture that it's like not everybody
is acting out the same exact script
and like you know um like not the same
exact movements whereas you look at
something like um you know uh opposite
end of of the the um the the political
representation maybe represented is like
you know um uh or or I'm trying to think
like historically uh like like um uh you
know um during during the Third Reich
it's like these were mixes of um of

internalist scripts of what you think
you're supposed to be acting out but
then there's also these strong scripts
involved where it's like you are
supposed to make exactly these motions
Etc Great points there and yes the pure
pressure I put it up there partially
because it so directly calls us to move
Beyond internalism and externalism I
mean what could be more external than
other people yet at the same time what
could be more internal psychologically
or neurophysiologically than your
interpretation of peer pressure and then
on the strong versus weak just like you
brought up um what could be weaker than
an invisible and lightly guiding force
in some cases like hey peer pressure is
like you should wear uh shoes that have
tied laces but no one's going to like
freak out but there's other peer
pressure things that people will freak
out and that's a strong script if 100%
of the time that something happens
socially enacted another thing happens
it's quite strong because 100% of the
time from State A it moves to State B
mediated by an external Force so that's
strong and external but there's other
cases where it's weak and internal at
least in how it appears and again the
goal of active inference isn't to be
like yep we judge strong versus weak and
it's strong it's rather to see this
Continuum as um an area of theoretical
dispute yes but now just a parameter in
our way of thinking about the world so
that there can be subcomponents or

different people in the same crowd who are experiencing very different realities or having very different um sort of generative statistics for their decision- making based upon some of these features we're discussing but yeah very interesting any other thoughts on the visualization otherwise there's um there's more we'll talk to because um it'll be good to have time for these two figures is that good all right let's go to these um two figures in the last half hour and just start out with just what are they showing because especially for those on the social sciences side it might be like whoo interesting conversation about the scripts and the chair and the being and the becoming but what are these variables and circles and lines so let's think about making that bridge between the qualitative things and the experiential and the social that we've been discussing with a more mathematical way that we're going to be returning to in all these different conversations so and as always people can just raise their hand but we're going to do maybe 10 to 15 minutes on each of these two figures so figure one in this paper is a simple generative model for policy selection and in act INF policy is always $P_i P$ for policy so this is what is being estimated the schematic depicts a generative model for policy selection it represents probabilistic beliefs about and then all of this is colorcoded so the first thing is a a is a

probabilistic belief about how observations are related to the states that cause them so here's observations this is like red Photon on my retina and this is state is like I'm wearing a red shirt okay so then there's this mapping in this case it's pretty simple where if there's a red shirt and the lights are on then I'm going to get red photons on my retina but one could imagine that there's State estimates that are social like this person is unhappy with me and there's some probab ability mapping onto different observations of body language or of um words for example so that's the a matrix it links the state estimates of the world to observations and a happens instantaneously in time now B is where we start taking lateral moves and this diagram goes from the left initial conditions to the right through the time steps 1 2 3 this is discrete time and B moving us from left to the right is mapping States onto each other so if we were talking about the red shirt well if you're wearing a red shirt at one moment basically 100% of the time you're wearing a red shirt at the next moment so that's one simple format or it could be again social like given my estimates of this person what do I estimate the transition Matrix is so knowing this person they get unhappy really fast but it's really easy to bring them back to being happy or they're always happy but once they get unhappy they're going to stay in that state for a while so you can imagine like a matrix where the on

and the off diagonals are different values and that captures whether once somebody is in a certain state is it an attractor state or a dwell state is it a state that they evacuate very rapidly like balancing on one foot so that's what's captured in this B Matrix is how things change through time um D is the initial conditions so those are states prior to sampling the world so that's like when we start having a conversation I am estimating your initial uh condition for that conversation um it's not that I know nothing about you I'm bringing in my estimate of before the conversation before I started sampling in this EPO where were you at and then in um the top we have G which includes C which is the preferences in the outcome matrices and so G that's definitely um uh unpacking equations for another day for 11 as well as for other discussions but let's just clarify this upper half G Bears on the selection of policy that's why in active inference we're so obsessed with free energy and how it relates to control theory cybernetics policy selection because all of the coolness of free energy is that it's a different way of thinking about how policy selection occurs so that's why G has an arrow pointing down to Pi now Pi is influencing only B's so this is what's pretty interesting is what is being estimated by the agent is like this whole graph okay but the connections within the model that are relevant are

not from policy to state so it's not saying how will States be different as a function of me with this investing strategy or of me with this social strategy or me with this navigation strategy how will the states be different actually it's policy to State transition matrices so this is like saying if I undertake a policy where I exercise every day then I believe that the the state transition Matrix from being unhealthy to being healthy will be modified not that I will move from unhealthy to healthy that's the state estimator that'd be policy to State and it does result in the same conclusion functionally but it's actually very very different from a mathematical and from a programmatic standpoint to say that the policy estimate Bears on how States transition into each other not just at the first level of what the states will be so again I can do policy one two 3 four five and instead of saying well if I do policy one then I would estimate that this state is most likely and if I do policy 2 this state is most likely that's the old way of thinking the B Matrix way of thinking is under policy one the world is apt to change in this way under policy 2 the world is apt to change in this way and then that is related back to affordances and preferences in the C Matrix as as well as other components of G that include for like the Curiosity and the maintenance of um rewarding States and all these other features so what this

graph is showing again just to sort of close out this little section is we're unpacking this idea of policy selection so just like if we were picking out how to drive from point A to point B we're unpacking this policy selection with an eye towards social interactions so where the state estimates are like whether the person is happy or unhappy with me what are the observations I'm going to see and how do those get mapped to each other what is their state through time or my state through time and then how does that change what were the initial conditions that set up the first observation we got and then as a function of a lot of cool stuff the sort of bulk of the iceberg in G how can policy be selected to shape the way that states transition to each other in a way that's most aligned with how I want the world to be and then agents that succeed and that succeed and those that don't don't so I hope that that sort of conveyed some of the attributes of this graph any thoughts or questions or what what's another context where this graph could be sort of lifted up and placed down

upon yeah so while people are thinking and raising their hand um again I would just try to give one more Bridge to the onboarding to active inference side because for those who are familiar with the formalism this is not even the full formalism they've probably already seen more expanded versions but for those for whom this type of diagram is relatively

novel in a social science setting just always keep in mind that no matter how many letters how many edges superscript subscript Greek letters you know cyc letters whatever there is we're trying to just be specific about what we're talking about and find a generalization or a structure that allows us to map structurally analogies from different social settings so this could be um so many different systems will fit under this type of model as well as and importantly that strong versus weak internal versus external idea that we were talking about earlier so we can have the initial conditions be me going outside with no shoes and here somebody else estimating um things about me or me estimating things about other people or this could be um the initial conditions of Hamlet in a given production or this could be the initial conditions of 2020 and so how in all of those different areas that qualitatively we were talking about how scripts could be used how are we going to find a sort of uh skeleton scaffold that all these other situations will be able to kind of hang on in a way that respects the specifics about their situation um as well as might even provide us with new knowledge so for some of you that might be hearing it twice the same way in a row for some of you it might be zero and maybe for others it's in the middle any thoughts though before we go to the next figure which is probably a bit

richer all right let's look at the next figure so anyways this was the skeleton generative model so hopefully we just familiarized ourself with the idea that things about social situations could be abstracted and represented with uh vectors and matrices and policy now we're going to take that to and here's in the paper where they write about how that could map now let's look at a little bit more expanded of an example okay so now we're looking at not just the agent but also the agent and the world or what are called here the internal realizer and the external realizer cuz in figure one one we were just looking at basically the internalist perspective we were just looking at you know you could imagine that a brain we're doing that or a computer we're doing that but there's nothing about the world there's no there's an estimate of the world state but there isn't an actual World state in this model so this is purely internal to the agent now let's look at figure two figure two is where we're going to start to bring in the world and we're going to bring it in at a very specific position and that is generating the observations that's really crucial the world doesn't intervene by jumping into the internal model now it might in reality this is the instrumentationalism of active inferences we're going to model it as if the only interface that the world could have is with observations and so in this figure one the only inputs to the model

were basically the initial conditions
and then the observables and now we're
picking right up here at the bottom of
the figure and we're going to graph the
world in so here's those observations
and those observations are being
generated by World states which are 11
so here reading from the um caption on
the side of the external world there's
only one process in play which is the
likelihood of observation States the
world 11 and the transitions between
these states in nine that's down here
okay so this from the point of view of
the agent is the generative process
that's creating the observations so like
there's a shirt that's generating the
red photons that is hitting my retina
for example but here's what's really
interesting is this model of hidden
State mapping observation transition
Matrix change in state emittance Matrix
observation again that is what we just
saw in figure
one flipped 180° State estimate and
transition Matrix and then leading to
observation
that is this deep and fundamental
symmetry that has been explored by axle
constant yell brunberg and others it's
this idea that we're going to do this
mathematical phrasing where like we know
which side we're on we know which side
is the internal side and which side is
enclosed by the epithelium but
mathematically look at
75857 and similarly 11 10 9 10 11 here
you just have have observation in the

middle it's on the interface that's like the retina being hit by the photon and then there's a mapping on the internal side from the agent through some type of likelihood Matrix to the observations so the agent is like reading it's trying to say what is the hidden State five given observation so what is the best seven to estimate five given the observation and then what is eight what's the transition Matrix from 5 to five and then similarly for the next time step and then on the other side it's the exact same except the key difference is that in the case of the internal agent we have six and so in six we have policy here it's a flirting policy but in the world there are things that change state to state in the world but we don't have policy control we don't have agency over those States and so then in order to select policy we have that g and that's kind of where we see the similarity to figure one so um it's just a very interesting formulation I'm sure each person's going to see their own thing in it and let's you know have some conversation or questions about uh what's unclear here but we're setting up the world and the agent theistic description of the generative model of the niche and of the agent we're setting them up in this intertwined and uh inextricable but also deeply symmetrical iCal way where if it was an agent who's like an organism and the niche is the forest so the forest doesn't enact any strategy it has a certain layout of acorns but that's not

a strategy that's like a chair for the squirrel and then the squirrel is the active inferring agent that's going to be doing things in this Forest so we can capture that here but also if it was squirrel versus squirrel combat and each squirrel was going to be having a generative model of the other squirrel having a generative model of it these are higher order theory of Mind recursive theory of mind it would be embedded in recursive because it'd be like about other active inference agents in the world and it could be deep it could be temporally deep it could include counterfactuals it could include affect or veilance that's what we'll be talking about in 11 but it's set up in this way that's very parsimonious because in the limit it allows for a true symmetry across the interface across the blanket in the Niche where the niche can be learning and making a model of the agent and vice versa so in the extreme it allows for that type of richness of agent agent Dynamics or this could be reduced to two inanimate objects so if there's two rocks that are just you know two bumps on a log and they're not doing free energy minimization or recursive estimation of anything they're just doing what they're doing with a certain State transition Matrix that can be captured within a reduced version of this framework so yeah not trying to rant on it but I just think that's kind of an interesting piece of it and we

often come back to this idea of Niche construction and it's like wait isn't control theory about what's the best strategy where is Niche construction coming into play internalism externalism it's like we're moving beyond internalism and externalism because we're saying you always need both you're always going to need both and there's going to be many types of interfaces one important interface Bas is seven and the observations which is like the sensory States the sensory cap capacities of the organism the Umwelt and the actual observations themselves that's like the handshake or the API call that actually crosses the interface those are critical but almost by definition because if there's something that doesn't cross the interface on the inbound side like a sense data that doesn't hit a neuron or on the outbound side like an action that doesn't act if it doesn't cross the interface and make the connection it's almost like it didn't happen and so we Define the interface in terms of the connections that do happen and then we can always be sure to be talking about relevant features of the world and always in a way that moves Beyond Simple conceptions of internalist versus externalist will have the ability to find situations where the internal features given a certain blanket of analysis dominate the Dynamics of the system or where given the inclusion of certain parameters or in a certain time scale of analysis the external features

dominate it but no one can say that's really a win for externalism or a win for internalism it's just a description of a situation so um I think it's an example about how we move Beyond some of these theoretical um and literature grounded

abstractions with a more abstract and general framework that actually lets us be more specific in the projects so we kind of escape this uncanny valley of whose team is better internalism or externalism we're going to escape that Valley by taking one step towards abstraction by like whoa what if we thought about the agent in the world as acting across an interface in the most General possible way but it turns out that by Framing it as an interface we're going to be able to be very very specific about the narratives of radicalization or the kinds of cues that are exchanged in a flirting relationship whereas if you're in the debate of internalism externalism you could have whole academic conferences on narratives of radicalization or behaviors of flirting and potentially the people would leave that meeting still feeling like their side had won so there wouldn't be a true conceptual synthesis so again a little bit another Spiral away another tangent but that's why these kinds of formalizations and specifically graphical and computable ways of thinking about them can bring Clarity to some of these long-standing qualitative questions and also

necessitate the collaboration of
basically quantitative and qualitative
expertises

RJ um so so um I I've seen like not
exactly this this graphic but but um uh
throughout each of the papers especially
the ones first's involved with there's
always some iteration of it um and some
iteration of explanation of active
inference and one thing um that
I've seen alluded to in the words but
not in the graphic is um this updating
of the observation itself um and maybe
this is being represented here and I'm
not seeing it um but but what what I
found is like um and and and I I'll get
to to why this is uh fairly important to
me in a second but but um it seems like
well in any process of sense making we
have this induction and at the state of
induction um you're actually like or
you're getting to trying to parse
meaning you're you're you're trying to
like check whether or not what you've
observed is even the correct observation
so it's the double take um and and this
is one of the reasons in intelligence
analysis that like collection is
separated from from from your your all
other elements of the production cycle
because that what they're trying to do
is they're trying to make sure that
there's not this like constant seduction
of going back and checking again
checking again checking again they're
trying to make it so there's a pipeline
so when something comes in it's going up
the pipe it's it's not your job to look

any further into whether or not what you saw was meant something or whether or not it was correct or it's just report what you saw push it through and now it's not your problem anymore it's up it's up somebody else's problem right um so like now at at the at at the the scale of the individual though um we're constantly double- checking ourselves whether or not we we saw we thought we saw so like if I'm if I'm reading a paper and I find um you know um something cited and and and I kind of know where that's coming from so this happened when we we um few people on this call we were working together on on on a paper for active inference and behavioral engineering and I was looking into a spirit decor and while I was doing that there was a big problem because there were so many different interpretations of what a spirit Decor meant so so what ended up happening is every time I saw the words I'm checking myself and going wait is am I actually observing what I think I'm observing I'm checking against priors and I'm updating my observation um so what what I'm kind of wondering is like um one like is this represented in this graphic and I'm not seeing it uh and then two um I know that over time we're starting to see okay like actual computational models um using active inference are those models like if they were applied in an agent-based model or applied in machine learning are those models trying to act that process out of like checking back

on the observation and making sure that
it's actually observing what it thinks
it's observing u_t rather than just
taking data in does that make sense yep
thanks for the question RJ so it is
lightly indicated in this figure as with
the other figure there's a movement from
left to right okay so the nodes are
numbered by the kind of things that they
are and so it's lightly indicated where
the direction of time is going with
these dotted arrows so like for example
the first 10 on the bottom left that's
like a hidden state of the world right
and through an 11 it emits an
observation at time point one then
there's a transition Matrix $\theta_{10,11}$ and so
if you did u_t hourly checkpoints the
 $\theta_{10,11}$ Matrix that you calculate is going
to be the hourly transition Matrix if
you do one minute checkpoints like a
discrete time with an interval of one
minute is going to be a different Matrix
so it's kind of like how the candle
chart of a stock or a crypto looks
different depending on what time length
the candle is so it's not the same
Matrix for all time intervals it's
always model interval dependent but $\theta_{10,11}$
moves us from the left 10 to the right
10 and then at that 10 again it's the
type 10 not like node number 10 it emits
through the same or a different 11
another observation now the early models
all the 11s were fixed so like the
hidden State could go from daytime to
night time but like the 11 was fixed and
then you could imagine that a more

advanced version of the model would be like yeah actually the hidden State through time it changes in how it emits 11 okay now flip the interface seven is perceiving the observation the left seven and then it's doing estimation of the hidden State s so that or five it just looks like an S and it is s in the models so it's kind of like at each time point we have this like 10 hidden state of the world emits through 11 in observation that's on the interface that's that holographic boundary that Chris Fields is going to help us understand then on the organism side the internalist side seven does a state estimate five then we move through time the organism updates its estimate through time with eight and the world updates its actual state with nine it's hidden State and then there's another time Point another stack of that and so in the example that you gave with like going back and double-checking information it's like you get an observation at one moment and then whether or not it's concordant with um something at least you're instantaneously sure of what it is but then here's another dimension that the computational models are going into which is at each time point it calculates an integral through time of policy so that's the part where we're going to talk about it in 11 and a few other ones but the simplest model would be like again red shirt or you know black shirt and then it's like the

photon is the observation and the state estimates pretty straightforward it's like a 2 x two Matrix there's an unchanging characters in The Matrix because the red shirt always emits red and there's no ambiguity okay now we can start to relax those assumptions and we can think about okay maybe there's more ambiguity in the signal maybe there's a non-stationary 11 Matrix maybe we can learn seven we can learn better about how observations are related to World States maybe we can learn eight better which is sort of like uh trying to reflect nine we can learn about how states in the world transition better and we can learn better policies too better sixes but that what we've just been focused on is really just the loop with the 10 n 10 n in the world 58 58 in the agent and then this interface that has a symmetry of the sevens and the 11s around the observation itself so yes they're the kinds of things that people are starting to roll out but I think that especially if you watch 11 coming up you'll get a sense at how rudimentary these rollouts are really it's at like the uh IBM chess 1995 status where like many many branches of counterfactuals are being processed and then evaluated and more modern chess computers don't use this Branch rollout evaluation strategy in the same way so we're still in a very early stage that's basically um defining what that roll out would be through time and then there will be other ways of heris approximating very

good

rollouts Sasha

yeah nice thanks for walking through

that on the diagram uh with all the

numbers um yeah uh just some things that

RJ reminded me of um is in thinking

about uh data collection uh it would be

great if we could separate um the data

collection from the data interpretation

and acting on it and um you know

informalized uh kind of scientific

approaches we can do that like um I'm

just thinking about my own work when I

do uh microscopy um and I'm looking at

some cells under a microscope and um I

might have like an emotional reaction to

how they look which is like they look

really good and I want to analyze them

or something went terribly wrong and

they look like um not even cells at this

point so um I think part of uh going

through this process is being able to

separate the individual steps so I have

to actively prevent myself from uh

yeah uh kind of throwing in the towel

preemptively when I don't uh that when

the data doesn't meet my expectations

until I go to the next step of like

randomizing and analyzing it um such

that I can remove my expectations um and

just try to collect what is there um and

I think that uh yeah that that's

something that's nearly impossible to do

in The Human Experience that we can't

really separate um the data collection

from the data interpretation and um yeah

it just moves too fast and then in the

um realm of like uncertainty and um yeah

what does this word mean in different contexts as R.J was saying that is the kind of thing that um in teaching like uh we try to think you know if you don't know every single word in what we're learning today um it's going to be really hard to build on any Next Level Concepts so and students will often say like well I don't really know this word so I don't know but the first thing to do is to like just look up what the word means so that you can move past that uncertainty and move on to move on to the next layers of things um and so that really to me makes me think of like social anxiety and uh people not being able to interpret social cues appropriately and that creating uncertainty which is just like it kind of snowballs once you're uncertain about one aspect of an interaction or a script um you kind of are prevented from participating in the next levels or carrying out the script and so that perhaps is something interesting to discuss um in next week's interact uh discussions cool thanks I think that question especially Sasha really makes me think about like script design and we're talking about the analysis and the descriptive ways of talking about systems but also about designing the scene and the stage um I think Steven dropped off but is somebody with uh drama experience performance experience you know social performance performing our roles a lot of these terms are starting to blend with the mask and

previously that would be seen as
dramatic or ritualized now it's seen as
a matter of public safety so many of
these things are actually coming
together in a way where we can um
certainly within our own life if not at
a higher level ask about what kind of
scripts we want to design for moving
forward especially about the uncertainty
from all these different issues so um
here's just a couple of questions I
wanted to throw up we only have you know
3 minutes so really just for the
listeners live or in replay to add
comments or start a discussion in
whatever platform we're using just
questions like thinking about how this
could be a unifying topic how could this
be a valuable model what unique insights
are we looking for how could we rethink
our experiences what would be
explanations or predictions and what
would be next steps for the research and
um thanks for participating I'm going to
just pause it for um on this final slide
as we hear our last thoughts so everyone
prepare a little last thought if you'd
like and to listeners thanks for
listening we will have follow-up forms
for the live participants it's in the
calendar invite and anyone is welcome to
provide us with feedback and stay in
touch but for the last 3 minutes let's
just hear from our panelists anybody who
would like to what is a last thought
what's something you're still wondering
about or now that we're done with 10
where does that leave us what are some

last thoughts R.J and then
blue uh so we I think we we did find
some conclusion on
um what can to use a script um but uh I
still don't feel like we have a good
conclusion on what a script isn't what
is that line so um I'm going to be
thinking about that the next week and I
hope we we touch on that next
discussion awesome
blue so I'm just wondering left really
wondering where can we use this to
engineer social behavior or to change
scripts that are perhaps unnecessary or
unwanted in society like radicalization
scripts or or things like that so that's
where I'm left you know how do we then
go out and apply what we've
learned cool um either of the Sasha or
Ivon um yeah I similar to Blue am left
thinking about what is the application
of active inference in scripts and um I
guess in instead of feeling uh uh
unsatisfied I I feel excited about it
that's my perspective um that it seems
that maybe at the end of the last
conversation I felt
uh ambivalent or confused about scripts
and now I'm a bit more uh directed in
their their use or their possibilities
so I'm I'm really happy to kind of have
uh talked through this and uh find a way
out that makes me feel excited about
scripts that hold space
specifically for things like freedom of
expression or speech or improvisation
it's like um a protection within the
script this is the part of the song

where there's a drum solo so for this part you get to play in time and then for this part it gets to be wacky and so in that way we can um and like blue is saying as well with the design we could be able to converge on scripts that protect certain freedoms as intrinsic to the structure of the script and being able to have a discussion about that that is um nuanced like we hopefully are representing and learning how to represent here that could be a really productive discussion instead of just about rights versus um all these other things that can get tangled down in so everybody thanks so much for participating and for the great discussion as always this is it for 10 we will have 11 in the final two weeks of December 2020 the 11th Hour the 11th act imp stream and then we will have 12 the clock will strike midnight on actm 2020 and we will be heading into a very exciting 2021 so thanks again for listening and for participating and see you later